

SIXTH SEMESTER (CUCBCSS—UG) DEGREE EXAMINATION, MARCH 2022**Mathematics****MAT 6B 13 (E01)—GRAPH THEORY****(2014 to 2018 Admissions)****(Multiple Choice Questions for SDE Candidates)****Time : 15 Minutes****Total No. of Questions : 20****Maximum : 20 Marks****INSTRUCTIONS TO THE CANDIDATE**

1. This Question Paper carries Multiple Choice Questions from 1 to 20.
2. The candidate should check that the question paper supplied to him/her contains all the 20 questions in serial order.
3. Each question is provided with choices (A), (B), (C) and (D) having one correct answer. Choose the correct answer and enter it in the main answer-book.
4. The MCQ question paper will be supplied after the completion of the descriptive examination.

MAT 6B 13 (E01)—GRAPH THEORY

(Multiple Choice Questions for SDE Candidates)

1. Let T be a tree. Then which of the following statements are true :
 - (A) There exists two paths between every pair of vertices.
 - (B) T is minimally connected.
 - (C) T has n vertices and $n - 2$ edges.
 - (D) T has n vertices and n edges.
2. Let $T = (V, E)$ be a tree. Then the eccentricity of the vertex is given by :
 - (A) $e(v) = \max \{d(u, v) : u \in V, u \neq v\}$.
 - (B) $e(v) = \min \{d(u, v) : u \in V, u \neq v\}$.
 - (C) $e(v) = \min \{d(u) : u \in V\}$.
 - (D) $e(v) = \max \{d(u) : u \in V\}$.
3. Let $G = (V, E)$ be a connected graph and let $v \in V$. Then v is a central point if :
 - (A) $e(v) > r(G)$.
 - (B) $e(v) < r(G)$.
 - (C) $e(v) = r(G)$.
 - (D) $e(v) \geq r(G)$.
4. Let $G = (V, E)$ be a graph such that $|E| = 8$ and $\deg(v) = 2$ for all $v \in V$. Then $|V| =$.
 - (A) 6.
 - (B) 2.
 - (C) 8.
 - (D) 10.
5. Let G_1 be a (p_1, q_1) graph and G_2 be a (p_2, q_2) graph. Then $G_1 + G_2$ is a :
 - (A) $(p_1 + p_2, q_1 + q_2)$ graph.
 - (B) $(p_1 q_1, p_2 q_2)$ graph.
 - (C) $(p_1 + p_2, q_1 + q_2 + p_1 p_2)$ graph.
 - (D) $(p_1 - p_2, q_1 - q_2)$ graph.
6. For what values of n is the graph K_n Eulerian ?
 - (A) n is an even number.
 - (B) n is an odd number.
 - (C) n is a composite number.
 - (D) None of these.
7. $K_{m,n}$ is Eulerian if and only if :
 - (A) m and n are even.
 - (B) m and n are odd.
 - (C) m is even and n is odd.
 - (D) m is odd and n is even.

8. Let $G_1 = (V_1, E_1)$ and $G_2 = (V_2, E_2)$ be two graphs. Let f be an isomorphism from G_1 to G_2 . Let $v \in V_1$. Then which of the following statements are true ?
- (A) $\deg(v) = \deg(f(v))$. (B) $\deg(v) > \deg(f(v))$.
 (C) $\deg(v) < \deg(f(v))$. (D) $\deg(v) \neq \deg(f(v))$.
9. The number of perfect matchings in the complete bipartite graph $K_{m,n}$ is :
- (A) n . (B) $n!$.
 (C) $(n-1)!$ (D) n^2 .
10. The number of perfect matching in K_{2n} is :
- (A) $2n!$. (B) $\frac{2n!}{n!}$.
 (C) $\frac{2n!}{2^n}$. (D) $\frac{2n!}{2^n n!}$.
11. Let G be a (p, q) plane graph in which every face is an n -cycle then $q =$
- (A) $\frac{n(p-2)}{n-2}$. (B) $\frac{(p-2)}{n-2}$.
 (C) $\frac{n(p+2)}{n-2}$. (D) $\frac{n(p-2)}{n+2}$.
12. Which of the following pairs of graphs are not planar ?
- (A) $K_2, K_{2,2}$. (B) $K_3, K_{2,2}$.
 (C) $K_4, K_{2,2}$. (D) $K_5, K_{3,3}$.
13. If a (p, q) graph is self dual, the $2p - q =$
- (A) 1. (B) 0.
 (C) 2. (D) 4.
14. Every block of a tree is :
- (A) K_2 . (B) K_3 .
 (C) K_4 . (D) K_5 .

15. Which of the following statements are true :

- (A) Every Hamiltonian graph is 0-connected.
- (B) Every Hamiltonian graph is 1-connected.
- (C) Every Hamiltonian graph is 2-connected.
- (D) Every Hamiltonian graph is 3-connected.

16. The value of $r(2, 2) =$

- (A) 0.
- (B) 1.
- (C) 2.
- (D) 3.

17. The graph $K_{r,s}$ is Hamiltonian if :

- (A) $r > s$.
- (B) $r < s$.
- (C) $r \neq s$.
- (D) $r = s$.

18. The chromatic number of $K_{m,n}$ is :

- (A) 1.
- (B) 2.
- (C) 3.
- (D) 4.

19. The chromatic number of C_{2n} is :

- (A) 1.
- (B) 2.
- (C) 3.
- (D) 4.

20. The value of $r(3, 3)$ is :

- (A) 0.
- (B) 1.
- (C) 4.
- (D) 6.

SIXTH SEMESTER (CUCBCSS-UG) DEGREE EXAMINATION, MARCH 2022

Mathematics

MAT 6B 13 (E01)—GRAPH THEORY

(2014 to 2018 Admissions)

Time : Three Hours

Maximum : 80 Marks

Section A

*Answer all questions.
Each question carries 1 mark.*

1. An edge joining a vertex to itself is called a _____.
2. Define simple graph.
3. Define complete graph.
4. Define Vertex degree.
5. Draw a 3-regular graph.
6. Define walk.
7. Define a tree.
8. Define bridge of a graph.
9. What is the number of different spanning trees of the complete graph K_n ?
10. Define cut vertex of a graph.
11. Define Hamiltonian cycle.
12. State Euler's formula for a plane graph.

(12 × 1 = 12 marks)

Section B

*Answer any nine questions.
Each question carries 2 marks.*

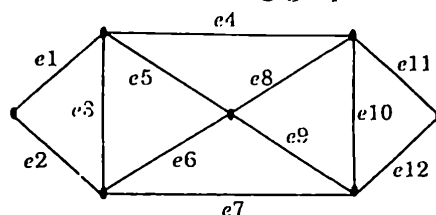
13. Define Graph.
14. For any graph with n vertices $v_1, v_2, \dots, v_n$ and ε edges, show that

$$\sum_{i=1}^n d(v_i) = 2\varepsilon.$$

15. Define complement of a graph. Give an example.

Turn over

16. Let G be a graph with n vertices and exactly $n - 1$ edges. Prove that G has either a vertex of degree 1 or an isolated vertex.
17. Draw the Petersen graph.
18. If G is an acyclic graph with k connected components, then show that G has $n - k$ edges.
19. If every edge of a graph G is a bridge, then show that G is a forest.
20. If G is an acyclic graph with $n - 1$ edges, then show that G is connected.
21. If v is a cut vertex of a graph G , then show that there are two vertices u and w of G , both different from v , such that v is on every $u - w$ path in G .
22. Give an Euler tour for the following graph :



23. Show that the dodecahedron is Hamiltonian.
24. Give an example for a planar graph and a non-planar graph.

(9 × 2 = 18 marks)

Section C

*Answer any six question.
Each question carries 5 marks.*

25. In any graph G , show that there is an even number of odd vertices.
26. Given any two vertices u and v of a graph G , show that every $u - v$ walk contains a $u - v$ path.

27. Draw the graph whose adjacency matrix is given by

$$\begin{bmatrix} 1 & 1 & 0 & 0 \\ 1 & 0 & 1 & 1 \\ 0 & 1 & 0 & 2 \\ 0 & 1 & 2 & 0 \end{bmatrix}.$$

28. Let G be a graph with n vertices $v_1, v_2, \dots, v_n$ and let A denote the adjacency matrix of G with respect to this listing of the vertices. Let $B = [b_{i,j}]$ be the matrix $B = A + A^2 + \dots + A^{n-1}$. Show that G is connected if and only if B has no zero entries off the main diagonal.

29. Let T be a tree with at least two vertices and let $P = u_0 u_1 \dots u_n$ be a longest path in T . Then show that both u_0 and u_n have degree 1.
30. If T is a tree with n vertices then show that it has precisely $n - 1$ edges.

31. Write a note on Kongsberg bridge problem.
32. Prove that it is impossible to have a group of nine people at a party such that each one known exactly five of the others in the group.
33. Let e be an edge of a connected graph G . Show that e is a bridge if and only if it is in every spanning tree of G .

(6 × 5 = 30 marks)

Section D

Answer any two question.

Each question carries 10 marks.

34. Let G be a non-empty graph with at least two vertices. Then show that G is bipartite if and only if it has no odd cycles.
35. (a) Show that a graph G is connected if and only if it has a spanning tree.
(b) Give an example of a non Hamiltonian connected graph.
36. (a) Let G be a graph in which the degree of every vertex is at least two, then show that G is acyclic.
(b) Prove that the wheel W_n is Hamiltonian for every $n \geq 4$.

(2 × 10 = 20 marks)